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Bachelor Thesis in Physics
submitted by

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February 28, 2026

Analysis of Cosmic Structure Formation under the Lens of Zel'dovich Flow and the Method of Characteristics

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Abstract

This thesis investigates cosmic structure formation using the Zel'dovich approximation and the method of characteristics, emphasizing the evolution of density perturbations and the onset of shell-crossing in a Lagrangian framework.

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Conclusion **43**

Motivation

> "Two things fill the mind with ever new and increasing admiration and awe, the more often and steadily we reflect upon them: the starry heavens above me and the moral law within me."
> — Immanuel Kant, *Critique of Practical Reason*

CHAPTER 1

Introduction

This thesis can broadly be split into three sections: Fourier analysis of the Vlasov equation under a perturbative Zel'dovich split. Then, the introduction of the method of characteristics as an approach to solving said Vlasov equation, followed by the analysis of how the mass probability function changes along a path slightly perturbed from the Zel'dovich trajectories.

CHAPTER 2

Mathematical and Theoretical Background

2.1 Cosmological Foundations

2.1.1 The expanding background and Zel'dovich coordinates

2.1.2 Einstein De Sitter Universe

Start with conformal time τ and comoving coords $\mathbf{x}$: $\mathbf{p} = am\,\frac{d\mathbf{x}}{d\tau} \implies \frac{d\mathbf{x}}{d\tau} = \frac{\mathbf{p}}{am}$.

$$\frac{df}{d\tau} = 0 \implies \frac{\partial f}{\partial \tau} + \frac{\partial f}{\partial \mathbf{x}} \cdot \frac{d\mathbf{x}}{d\tau} + \frac{\partial f}{\partial \mathbf{p}} \cdot \frac{d\mathbf{p}}{d\tau} = 0 \tag{2.1}$$

Since we are working with the canonical momentum, we can extract the first derivative of the momentum from the Hamilton equation:

$$\dot{\mathbf{p}} = -\frac{\partial \mathcal{H}}{\partial \mathbf{q}}, \quad \text{where } q = x \text{ and } \mathcal{H}(p,x) = \frac{p^2}{2m} + \underbrace{V(x)}_{=\tilde{\Phi}} \tag{2.2}$$

$$\implies \frac{d\mathbf{p}}{d\tau} = -am\,\nabla_x \tilde{\Phi} \implies \frac{df}{d\tau} = \frac{\partial f}{\partial \tau} + \frac{1}{am}\nabla_x f - am\,\nabla_x \tilde{\Phi}\,\nabla_p f = 0 \tag{2.3}$$

where

$$\Delta \tilde{\Phi}(\mathbf{x},\tau) = \frac{3}{2}\,\Omega_m(\tau)\,H^2(\tau)\,\delta(\mathbf{x},\tau) \tag{2.4}$$

Furthermore in comoving coords we have:

$$\ddot{\mathbf{q}} + H\dot{\mathbf{q}} = -\nabla_{\mathbf{q}}\tilde{\Phi} \quad (*) \tag{2.5}$$

$\implies$ we introduce $\eta = \log\left(\frac{D(\tau)}{D^{(i)}}\right)$ and the non-canonical momentum $\mathbf{p} = m\,\frac{d\mathbf{x}}{d\eta}$.

$$\implies \frac{df}{d\eta} = \frac{\partial f}{\partial \eta} + \frac{\partial f}{\partial \mathbf{x}} \cdot \frac{d\mathbf{x}}{d\eta} + \frac{\partial f}{\partial \mathbf{p}} \cdot \frac{d\mathbf{p}}{d\eta} = 0 \tag{2.6}$$

$$\boxed{\begin{aligned}&f_+ = \tfrac{d\log b_+}{d\log a},\ H = \tfrac{\dot a}{a};\ \eta = \log\left(\tfrac{D(\tau)}{D^{(i)}}\right)\\ &= \tfrac{\dot D_+}{D_+}\cdot\tfrac{a}{\dot a} = \tfrac{d\eta}{d\tau}\cdot\tfrac{1}{H}\\ &\implies \tfrac{d\eta}{d\tau} = f_+ H\end{aligned}}$$

$$\dot{\mathbf{q}} = \dot{\mathbf{x}} = \frac{d\mathbf{x}}{d\tau} = \frac{\partial\mathbf{x}}{\partial\eta}\cdot\frac{d\eta}{d\tau} = \frac{\mathbf{p}}{m}\cdot f_+ H \tag{2.7}$$

$$\ddot{\mathbf{q}} = \frac{1}{m}\left[(f_+H)^2\frac{d\mathbf{p}}{d\eta} + \mathbf{p}\frac{d}{d\tau}(f_+H)\right] \tag{2.8}$$

$$= \frac{1}{m}\left[(f_+H)^2\frac{d\mathbf{p}}{d\eta} + \mathbf{p}\frac{d}{d\tau}(f_+H)\right] \tag{2.9}$$

From linear growth eq.:

$$\ddot D + H\dot D = \frac{3}{2}\Omega_m H^2 D \implies \frac{\ddot D}{D} + H\frac{\dot D}{D} = \frac{3}{2}\Omega_m H^2 \tag{2.10}$$

$$\implies \frac{d}{d\tau}(f_+H) = \frac{\ddot D}{D} - \left(\frac{\dot D}{D}\right)^2 = \frac{3}{2}\Omega_m H^2 - f_+H^2 - (f_+H)^2 \tag{2.11}$$

Inserting into $(*)$ we get

$$\frac{1}{m}\left[(f_+H)^2\frac{d\mathbf{p}}{d\eta} + \mathbf{p}\frac{d}{d\tau}(f_+H)\right] + \frac{\mathbf{p}}{m}f_+H^2 = -\nabla_{\mathbf{q}}\tilde\Phi \tag{2.12}$$

$$\implies \frac{d\mathbf{p}}{d\eta} = -\mathbf{p}\left[\frac{3}{2}\frac{\Omega_m}{f_+} - 1\right] - \nabla_{\mathbf{q}}\tilde\Phi\cdot m \approx -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde\Phi \tag{2.13}$$

using the EdS approximation: $\Omega_m = f_+ = 1$, and $\tilde{\tilde\Phi} = \tilde\Phi/(Hf_+)^2$.

$$\boxed{\frac{\partial f}{\partial\eta} + \frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{q}}f - \left[\frac{\mathbf{p}}{2} + m\,\nabla_{\mathbf{q}}\tilde{\tilde\Phi}\right]\cdot\nabla_{\mathbf{p}}f = 0} \tag{2.14}$$

2.2 Klimontovich equation and the BBGKY Hierarchy

$$\frac{d}{dt}\vec q_j(t) = \vec\nabla_{\vec p_j}H(x_j,t), \qquad \frac{d}{dt}\vec p_j(t) = -\vec\nabla_{\vec q_j}H(x_j,t), \qquad j = 1,\dots,N \tag{2.1}$$

$$\dot x_j(t) = \omega\,\nabla_{x_j}H\Big(x_j(t),t\Big), \qquad \omega = \begin{pmatrix} 0 & I_{3\times3}\\ -I_{3\times3} & 0\end{pmatrix} \tag{2.2}$$

$$H(x,t) = H_0 + H_{\text{int}}(x,t) \tag{2.3}$$

$$= \frac{1}{2}x^{\mathrm T}A(t)x + \frac{1}{2}\sum_{i=1}^{N}v_1(x_i,t) + \frac{1}{2}\sum_{\substack{i,j=1\\ i\neq j}}^{N}v_2(x_i,x_j,t) \tag{2.4}$$

where $A(t)$ is some symmetric matrix and the v_1 term represents external field contributions and v_2 pairwise interactions.

$$\dot{x}^{(0)}(t) = \omega \cdot A(t) \cdot x^{(0)}(t) \tag{2.5}$$

$$x_{\text{cl}}^{(0)}(t) = G(t, t^{(i)}) \cdot x^{(i)} \tag{2.6}$$

$$G(t,t') = \begin{pmatrix} g_{qq}(t,t')I_{3\times3} & g_{qp}(t,t')I_{3\times3} \\ g_{pq}(t,t')I_{3\times3} & g_{pp}(t,t')I_{3\times3} \end{pmatrix} \Theta(t-t') \tag{2.7}$$

$$f_1(x,t) = \langle \Phi_f(x,t) \rangle = \int dx \sum_{i=1}^{N} \delta_D(x - x_i(t)) \rho_N(x,t) \tag{2.8}$$

$$\Phi_f(x,t) = \sum_{i=1}^{N} \delta_D(x - x_i(t)) \tag{2.9}$$

$$\int d^6x \, f_1(x,t) = N \tag{2.10}$$

$$\bar{C}_{ff}(x_1,x_2,t) = \langle \Phi_f(x_1,t)\Phi_f(x_2,t) \rangle = \delta_D(x_1 - x_2) f_1(x_1,t) + f_2(x_1,x_2,t) \tag{2.11}$$

$$f_2(x_1,x_2,t) = f_1(x_1,t) f_1(x_2,t) + g_2(x_1,x_2,t) \tag{2.12}$$

$$\bar{G}_{ff}(x_1,x_2,t) = \langle \Phi_f(x_1,t)\Phi_f(x_2,t) \rangle_c = \delta_D(x_1 - x_2) f_1(x_1,t) + g_2(x_1,x_2,t) \tag{2.13}$$

$$f_s(x_1,\ldots,x_s,t) := \left\langle \sum_{\substack{i_1,\ldots,i_s=1 \\ i_1 \neq \cdots \neq i_s}} \prod_{n=1}^{s} \delta_D(x_n - x_{i_n}(t)) \right\rangle \tag{2.14}$$

$$\int d^6x_1 \cdots d^6x_s \, f_s(x_1,\ldots,x_s,t) = \frac{N!}{(N-s)!} \tag{2.15}$$

$$f_s(x_1,\ldots,x_s,t) = \prod_{n=1}^{s} f_1(x_n,t) \tag{2.16}$$

$$\bar{C}_{f\ldots f}(x_1,\ldots,x_k,t) := \langle \Phi_f(x_1,t) \cdots \Phi_f(x_k,t) \rangle \tag{2.17}$$

$$= \int dx \prod_{i=1}^{k} \sum_{i_n=1}^{N} \delta_D(x_i - x_i(t)) \rho_N(x,t) \tag{2.18}$$

$$\frac{\partial}{\partial t_1} \Phi_f(x_1,t_1) = \sum_{i=1}^{N} \nabla_{x_i}^T \delta_D(x_1 - x_i(t)) \cdot \omega \cdot \nabla_{x_i} H(x,t) \tag{2.19}$$

using (2.4) we get

$$\left[\frac{\partial}{\partial t_1} - \hat{\mathcal{L}}_1^{(0)}(t_1) - \hat{\mathcal{L}}_1^{(1)}(t_1) \right] \Phi_f(x_1,t_1) = \int d^6x_2 \hat{\mathcal{L}}_{12}^{(2)}(t_1) \Phi_f(x_1,t_1) \Phi_f(x_2,t_1) \tag{2.20}$$

where we have the free evolution operator $\hat{\mathcal{L}}_1^{(0)}(t_1)$, the one particle operator which represents external field contributions $\hat{\mathcal{L}}_1^{(1)}(t_1)$ and the two particle operator for pairwise interactions $\hat{\mathcal{L}}_{12}^{(2)}(t_1)$

$$\hat{\mathcal{L}}_1^{(0)}(t_1) = x_1^T \cdot A(t_1) \cdot \omega \cdot \nabla_{x_1} \tag{2.21}$$
$$\hat{\mathcal{L}}_1^{(1)}(t_1) = \nabla_{x_1}^T v_1(x_1, t_1) \cdot \omega \cdot \nabla_{x_1} \tag{2.22}$$
$$\hat{\mathcal{L}}_{12}^{(2)}(t_1) = \nabla_{x_1}^T v_2(x_1, x_2, t_1) \cdot \omega \cdot \nabla_{x_1} \tag{2.23}$$

We can generalise this to s particle systems, taking the Liuville equation for the s-particle phase space density yields the BBGKY hierarchy

$$\left[\frac{\partial}{\partial t_1} - \sum_{i=1}^{s} \left(\hat{\mathcal{L}}_i^{(0)}(t_1) + \hat{\mathcal{L}}_i^{(1)}(t_1)\right)\right] f_s(x_1, \ldots, x_s, t_1) \tag{2.24}$$
$$= \int d^6 x_{s+1} \sum_{i=1}^{s} \hat{\mathcal{L}}_{i,s+1}^{(2)}(t_1) f_{s+1}(x_1, \ldots, x_{s+1}, t_1) \tag{2.25}$$

2.3 Vlasov Equation

In the BBGKY Hierarchy, we notice that knowledge on the evolution of the s particle distribution function requires knowledge of the (s+1) particle distribution function. To close this hierarchy, we must find a way to truncate the equation in a way that we can solve it.
We can take the free streaming ansatz, which assumes no pairwise interactions between particles:

$$v_2(x_1, x_2, t) \to \lambda v_2(x_1, x_2, t) \tag{2.1}$$
$$f_1(x_1, t) \sim \Theta(\lambda^{(0)}) \tag{2.2}$$
$$g_2(x_1, x_2, t) \sim \Theta(\lambda^{(1)}) \tag{2.3}$$

we set $\lambda \to 0$ and neglect all terms of order $\Theta(\lambda^{(1)})$ and higher, yielding:

$$\left[\frac{\partial}{\partial t_1} - \hat{\mathcal{L}}_i^{(0)}(t_1)\right] f_1(x_1, t_1) = 0 \tag{2.4}$$
$$f_1(x_1, t_1 = t^{(i)}) =: f^{(i)}(x_1) \tag{2.5}$$

we introduce the retarder Greens function:

$$D_{0,R}(x_1, t_1, x_2, t_2) = \delta_D(x_1 - G(t_1, t_2)x_2)\Theta(t_1 - t_2) \tag{2.6}$$
$$\Rightarrow \left[\partial_{t_1} - \hat{\mathcal{L}}_1^{(0)}(t_1)\right] D_{0,R}(x_1, t_1, x_2, t_2) = \delta_D(x_1 - x_2)\delta_D(t_1 - t_2) \tag{2.7}$$
$$\tag{2.8}$$

from which it follows:

$$f_1(x_1, t) = \int d^6 x_2 D_{0,R}(x_1, t, x_2, t^{(i)}) f^{(i)}(x_2) \tag{2.9}$$
$$= f^{(i)}(G^{-1}(t, t^{(i)})x_1)\Theta(t_1 - t^{(i)}) \tag{2.10}$$

including the next order in coupling leads to the Vlasov equation:

$$\left[\frac{\partial}{\partial t_1} - \hat{\mathcal{L}}_1^{(0)}(t_1) - \hat{\mathcal{L}}_1^{(1)}(t_1)\right] f_1(x_1,t_1) = \int d^6x_2 \hat{\mathcal{L}}_{12}^{(2)}(t_1) f_1(x_1,t_1) f_1(x_2,t_1) \tag{2.11}$$

2.4 Momentum Moments and Closure

Let $f_s(\mathbf{q},\mathbf{p},\tau)$ be the full 6D phase-space density. We define the momentum moments

$$n(\mathbf{q},\tau) := \int d\mathbf{p}\; f_s(\mathbf{q},\mathbf{p},\tau), \tag{2.1}$$

$$\Pi(\mathbf{q},\tau) := \int d\mathbf{p}\; \mathbf{p}\; f_s(\mathbf{q},\mathbf{p},\tau) = am\, n(\mathbf{q},\tau)\; \mathbf{u}(\mathbf{q},\tau), \tag{2.2}$$

$$T(\mathbf{q},\tau) := \int d\mathbf{p}\; \mathbf{p}\otimes\mathbf{p}\; f_s(\mathbf{q},\mathbf{p},\tau) = a^2m^2 n(\mathbf{q},\tau)\Big(\mathbf{u}(\mathbf{q},\tau)\otimes\mathbf{u}(\mathbf{q},\tau) + \sigma(\mathbf{q},\tau)\Big), \tag{2.3}$$

where the bulk velocity and velocity-dispersion tensor are

$$u_i(\mathbf{q},\tau) = \frac{1}{am\, n(\mathbf{q},\tau)} \int d\mathbf{p}\; p_i\; f_s(\mathbf{q},\mathbf{p},\tau), \tag{2.4}$$

$$\sigma_{ij}(\mathbf{q},\tau) = \frac{1}{a^2m^2 n(\mathbf{q},\tau)} \int d\mathbf{p}\; \Big(p_i - am\, u_i(\mathbf{q},\tau)\Big)\Big(p_j - am\, u_j(\mathbf{q},\tau)\Big)\; f_s(\mathbf{q},\mathbf{p},\tau). \tag{2.5}$$

We describe the new degrees of freedom in a way that avoids redundancy $\rightarrow$ *cumulants*. Define the characteristic (moment-generating) function

$$Z(\mathbf{q},\mathbf{l},\tau) = \int d\mathbf{p}\; e^{i\mathbf{l}\cdot\mathbf{p}}\; f_s(\mathbf{q},\mathbf{p},\tau), \tag{2.6}$$

with moments

$$M^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau) = i^n \left.\frac{\partial^n Z(\mathbf{q},\mathbf{l},\tau)}{\partial l_{i_1}\cdots\partial l_{i_n}}\right|_{\mathbf{l}=0}, \tag{2.7}$$

so that (expansion around $\mathbf{l}=0$)

$$Z(\mathbf{q},\mathbf{l},\tau) = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} M^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau)\; l_{i_1}\cdots l_{i_n}. \tag{2.8}$$

The cumulant-generating function and its cumulants are

$$C(\mathbf{q},\mathbf{l},\tau) := \ln Z(\mathbf{q},\mathbf{l},\tau) = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} C^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau)\; l_{i_1}\cdots l_{i_n}, \tag{2.9}$$

$$C^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau) = i^n \left.\frac{\partial^n C(\mathbf{q},\mathbf{l},\tau)}{\partial l_{i_1}\cdots\partial l_{i_n}}\right|_{\mathbf{l}=0}. \tag{2.10}$$

The first three cumulants identify with the physical moments:

$$C^{(0)}(\mathbf{q},\tau) = \ln n(\mathbf{q},\tau), \tag{2.11}$$

$$C^{(1)}_i(\mathbf{q},\tau) = am\, u_i(\mathbf{q},\tau), \tag{2.12}$$

$$C^{(2)}_{ij}(\mathbf{q},\tau) = a^2m^2\, \sigma_{ij}(\mathbf{q},\tau). \tag{2.13}$$

The full distribution can therefore be written as

$$Z(\mathbf{q},\mathbf{l},\tau) = \exp\left(\sum_{n=0}^{\infty} \frac{(-i)^n}{n!} C^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau)\, l_{i_1}\cdots l_{i_n}\right), \tag{2.14}$$

$$f_s(\mathbf{q},\mathbf{p},\tau) = \int \frac{d\mathbf{l}}{(2\pi)^3}\, e^{i\mathbf{l}\cdot\mathbf{p}} \exp\left(\sum_{n=0}^{\infty} \frac{(-i)^n}{n!} C^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau)\, l_{i_1}\cdots l_{i_n}\right). \tag{2.15}$$

2.5 SSA: Single Stream Approximation

- Set all cumulants of orders higher than 1 to zero.
- $f_s(\mathbf{q},\mathbf{p},\tau)$ is an infinitely thin 3D sheet in 6D phase space $\rightarrow$ the system is in a perfectly cold state.

$$f_s(\mathbf{q},\mathbf{p},\tau) = n(\mathbf{q},\tau) \int \frac{d\mathbf{l}}{(2\pi)^3} \exp\Big(i\mathbf{l}\cdot(\mathbf{p} - am\,\mathbf{u}(\mathbf{q},\tau))\Big) = n(\mathbf{q},\tau)\, \delta^{(3)}\Big(\mathbf{p} - am\,\mathbf{u}(\mathbf{q},\tau)\Big). \tag{2.1}$$

2.5.1 Gaussian Closure

- Non-vanishing velocity dispersion.
- Set all cumulants of order higher than 2 to zero.

$$\Rightarrow \quad f_s(\mathbf{q},\mathbf{p},\tau) = n(\mathbf{q},\tau) \int \frac{d\mathbf{l}}{(2\pi)^3} \exp\Big(i\mathbf{l}\cdot(\mathbf{p} - am\,\mathbf{u}(\mathbf{q},\tau)) - \tfrac{1}{2}\sigma_{ij}\, l_i l_j\Big). \tag{2.2}$$

We use the standard Gaussian integral identity: for symmetric, positive definite (and invertible) matrices $A \in \mathbb{R}^{n\times n}$ and $\mathbf{b} \in \mathbb{R}^n$,

$$\int_{\mathbb{R}^n} d^n\mathbf{x}\, \exp\Big(-\tfrac{1}{2}\mathbf{x}^T A\mathbf{x} + \mathbf{b}^T\mathbf{x}\Big) = \frac{(2\pi)^{n/2}}{\sqrt{\det A}}\, \exp\Big(\tfrac{1}{2}\mathbf{b}^T A^{-1}\mathbf{b}\Big). \tag{2.3}$$

The velocity dispersion σ_{ij} is symmetric by definition. We must also require it to be *positive definite*:

$$V_i\, \sigma^{ij}\, V_j = \frac{1}{a^2 m^2 n} \int d\mathbf{p}\, V_i(p_i - am\, u_i)\, V_j(p_j - am\, u_j)\, f_s = \frac{1}{a^2 m^2}\, \mathrm{Var}(\mathbf{V}\cdot\mathbf{p}). \tag{2.4}$$

Positive definiteness requires $V_i\sigma^{ij}V_j > 0$, i.e., $\mathrm{Var}(\mathbf{V}\cdot\mathbf{p}) > 0$: no vanishing support in any direction. This is already guaranteed by the constraint of isotropy. ✓
Identifying $\mathbf{b} = i(\mathbf{p} - am\,\mathbf{u}(\mathbf{q},\tau))$, we obtain the *Maxwell–Boltzmann form* of the Gaussian closure:

$$\boxed{f_s(\mathbf{q},\mathbf{p},\tau) = \frac{n(\mathbf{q},\tau)}{(2\pi)^{3/2}\sqrt{\det\sigma}} \exp\Big(-\frac{1}{2}(p_i - am\, u_i)(\sigma^{-1})_{ij}(p_j - am\, u_j)\Big).} \tag{2.5}$$

2.6 Fourier transformation of the Vlasov equation

CHAPTER 3

Zel'dovich Split

In this section our goal is to solve the Vlasov equation in Fourier space under a perturbative expansion around the Zel'dovich solution. From this section onwards f will be referring to the Fourier transformed one particle distribution function unless otherwise specified.
We start by writing down the Vlasov equation in operator form:

$$\mathcal{L}_0 f + \mathcal{L}_{Grav} f = 0 \tag{3.1}$$

$$\rightarrow \mathcal{L}_0 \left[f_{0,Zel} + \delta f\right] + \mathcal{L}_{Grav} \left[f_{0,Zel} + \delta f\right] = 0 \tag{3.2}$$

We know by construction that the Zel'dovich background $f_{0,Zel}$ follows the Zel'dovich flow, which reproduces linear growth, so by definition the Zel'dovich background satisfies:

$$\mathcal{L}_0 f_{0,Zel} + \mathcal{L}^{(1)}_{Grav} f_{0,Zel} = 0 \tag{3.3}$$

, where $\mathcal{L}^{(1)}_{Grav}$ denotes the first order gravitational interaction operator.

$$\mathcal{L}_{Grav} = \mathcal{L}^{(1)}_{Grav} + \mathcal{L}^{(2)}_{Grav} + ... \tag{3.4}$$

Knowing this we can factor out (3.2) and (3.3) leaves us with:

$$\mathcal{L}_0 \delta f + \mathcal{L}^{(rest)}_{Grav} \delta f + \mathcal{L}_{Grav} f_{0,Zel} = 0 \tag{3.5}$$

, where

$$\mathcal{L}^{(rest)}_{Grav} = \mathcal{L}_{Grav} - \mathcal{L}^{(1)}_{Grav} = \sum_{n \geq 2} \mathcal{L}^{(n)}_{Grav} \tag{3.6}$$

We can further define the Zel'dovich operator as:

$$\mathcal{L}_{Zel} := \mathcal{L}_0 + \mathcal{L}^{(1)}_{Grav} = G_Z^{-1} \tag{3.7}$$

also known as the Zel'dovich 'free' propagator.

$$\mathcal{L}_0\delta f + \mathcal{L}_{Grav}\delta f = -\sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\delta f \tag{3.8}$$

$$\Leftrightarrow \mathcal{L}_0\delta f + \mathcal{L}^{(1)}_{Grav}\delta f + \sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\delta f = -\sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\delta f \tag{3.9}$$

$$\Leftrightarrow \mathcal{L}_{Zel}\delta f = -\sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\left[f_{0,Zel} + \delta f\right] \tag{3.10}$$

$$\tag{3.11}$$

From here we can solve for δf using the Zel'dovich propagator/ Greens function:

$$\delta f = G_Z(\eta,\eta')\delta f^{(i)}(\eta') - \int d\eta' G_Z(\eta,\eta')\sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\left[f_{0,Zel} + \delta f\right] \tag{3.12}$$

while we now have an expression for the perturbations, we still need to explicitly define the higher order gravitational operator. From 2.6 we have that the gravitational operator in Fourier space is given by:

$$\mathcal{L}_{Grav}f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\left[f_{0,Zel}(k-k',l,\eta) + \delta_Z f(k-k',l,\eta)\right]\left[f_{0,Zel}(k',0,\eta) + \delta_Z f(k',0,\eta)\right] \tag{3.13}$$

we now introduce the $f_{0,Zel} = f_{0,com} + \delta f^{(1)}$, where $(\cdot)^1$ now denotes the first order perturbations with respect to the comoving coordinates and

$$\delta_Z f = \delta f^{(2)} + \delta f^{(3)} + ... \tag{3.14}$$

furthermore:

$$f_{0,Zel}(k-k',l,\eta) \equiv f_{Z,0} \quad f_{0,Zel}(k',0,\eta) \equiv f^{\Phi}_{Z,0} \tag{3.15}$$

$$\Rightarrow \quad \mathcal{L}_{\text{Grav}}f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\left[f_{Z,0} + \delta_Z f_Z\right]\left[f^{\Phi}_{Z,0} + \delta_Z f^{\Phi}_Z\right] \tag{3.16}$$

$$= -\frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\Big[f_{Z,0}f^{\Phi}_{Z,0} + f^{\Phi}_{Z,0}\delta_Z f_Z + f_{Z,0}\delta_Z f^{\Phi}_Z + \delta_Z f^{\Phi}_Z\delta_Z f_Z\Big] \tag{3.17}$$

furthermore we know that:

$$f_{Z,0} = f_{0,com} + \delta f^{(1)} \tag{3.18}$$

$$\begin{aligned}\mathcal{L}_{\text{Grav}}f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\Big[&(f_{0,\text{com}} + \delta f^{(1)})(f^{\Phi}_{0,\text{com}} + \delta f^{(1)\Phi}) \\ &+ (f^{\Phi}_{0,\text{com}} + \delta f^{(1)\Phi})\,\delta_Z f_Z + (f_{0,\text{com}} + \delta f^{(1)})\,\delta_Z f^{\Phi}_Z + (\delta_Z f^{\Phi}_Z)(\delta_Z f_Z)\Big]\end{aligned} \tag{3.19}$$

factoring out lets us separate $\mathcal{L}_{Grav}$ in contributions of perturbation, where $f_{0,com}$ is of order 0, $\delta f^{(1)}$ of order 1 and δ_Z denotes perturbations of order 2 and higher.

$$\Rightarrow \quad \mathcal{L}^{(0)}_{\text{Grav}} f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\, f_{0,\text{com}} f^{\Phi}_{0,\text{com}} = 0 \tag{3.20}$$

$$\mathcal{L}^{(1)}_{\text{Grav}} f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\left[f_{0,\text{com}}\,\delta f^{(1)\Phi} + f^{\Phi}_{0,\text{com}}\,\delta f^{(1)}\right] \tag{3.21}$$

$$\sum_{n\geq 2}\mathcal{L}^{(n)}_{\text{Grav}} f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\Big[\delta f^{(1)}\delta f^{(1)\Phi} + f_{0,\text{com}}\,\delta_Z f^{\Phi}_Z + \delta f^{(1)}\delta_Z f^{\Phi}_Z \tag{3.22}$$

$$+ f^{\Phi}_{0,\text{com}}\,\delta_Z f_Z + \delta f^{(1)\Phi}\delta_Z f_Z + \delta_Z f^{\Phi}_Z \delta_Z f_Z\Big] \tag{3.23}$$

Inserting this back into (3.12) yields an expression for the perturbations δf around the Zel'dovich background:

$$\delta f^{(n)}(k,l,\eta) = \int \frac{d^3q_1}{(2\pi)^3}\cdots\frac{d^3q_n}{(2\pi)^3}\delta^{(i)}(q_1)\ldots\delta^{(i)}(q_n)F_n(k,l;q_1,\ldots,q_n,\eta) \tag{3.24}$$

for now we will leave this section as it is and move on to the method of characteristics.

CHAPTER 4

Method of Characteristics

4.1 Fourier space characteristics

We start by writing down the Vlasov equation in Fourier space again:

$$\left[\frac{\partial}{\partial\eta} - k\cdot\nabla_l + \frac{1}{2}l\cdot\nabla_l\right] f(k,l,\eta) - \frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'l}{k'^2} f(k-k',l,\eta) f(k',0,\eta) = 0 \tag{4.1}$$

on our way to apply the method of characteristics to the Zel'dovich split Vlasov equation we take a small detour through some simpler cases as a proof of concept.

4.1.1 Free-Streaming ansatz

The free streaming ansatz ignores all gravitational interactions such that we can write:

$$\left[\frac{\partial}{\partial\eta} - k\cdot\nabla_l + \frac{1}{2}l\cdot\nabla_l\right] f(k,l,\eta) = 0 \tag{4.2}$$

we now wish to find trajectories $(\eta(s), k(s), l(s))$ such that along these paths we get an expression:

$$\frac{\partial}{\partial s} f(k(s), l(s), \eta(s)) = F(s, f(k(s), l(s), \eta(s))) \tag{4.3}$$

we can start by taking the derivative of f with respect to this affine parameter s and using the chain rule yields:

$$\frac{d}{ds} f = \frac{\partial f}{\partial s}\frac{d\eta}{ds} + \frac{\partial f}{\partial k}\frac{dk}{ds} + \frac{\partial f}{\partial l}\frac{dl}{ds} = F \tag{4.4}$$

comparing this to (4.2) we can identify:

$$\frac{d\eta}{ds} = 1 \tag{4.5}$$
$$\frac{dk}{ds} = 0 \tag{4.6}$$
$$\frac{dl}{ds} = -k + \frac{1}{2}l \tag{4.7}$$
$$F = 0 \tag{4.8}$$

we can justify setting $\eta = s$ by stating that $s(0) = 0$ and from these equations it follows that:

$$k(\eta) = k_0 := k(0) \tag{4.9}$$
$$l(\eta) = 2k + Ce^{\eta/2} \quad \Rightarrow \quad l(\eta) = 2k_0 + (l_0 - 2k_0)e^{\eta/2} \tag{4.10}$$

where the final step results from setting $l(0) = l_0$.
This specific case can be seen as the trivial case, where the distribution function remains unchanged along the characteristics

$$\frac{df}{ds} = 0 \quad \Rightarrow \quad f(\eta, k, l) = f_0 \tag{4.11}$$

4.1.2 Zel'dovich 'free' streaming

Before going on with this section it is perhaps important to clarify the quotation marks around the word 'free'. In this subsection we will only be solving for the evolution function generated along the trajectories of the characteristics given by solely by the Zel'dovich operator (3.7), which includes first order gravitational interactions and thus cannot in good faith be called a proper free streaming ansatz. However, for the sake of simplicity we will be using this naming convention henceforth.

$$\left[\frac{\partial}{\partial\eta} - k\cdot\nabla_l + \frac{1}{2}l\cdot\nabla_l\right](f_{0,com} + \delta f^{(1)}) - \frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'l}{k'^2}\left[f_{0,com}\delta f^{(1)\Phi} + f^{\Phi}_{0,com}\delta f^{(1)}\right] = 0 \tag{4.12}$$

$$\Rightarrow \left[\frac{\partial}{\partial\eta} - k\cdot\nabla_l + \frac{1}{2}l\cdot\nabla_l\right]\delta f^{(1)} - \frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'l}{k'^2}\left[f_{0,com}\delta f^{(1)\Phi} + f^{\Phi}_{0,com}\delta f^{(1)}\right] = 0 \tag{4.13}$$

where the background term drops out, as this is the free streaming case solved in the previous subsection.
What we can notice here that $\delta f^{(1)}$ determines the entirety of the dynamics and the characteristics. One should also note the characteristics are the exact same as in the free streaming approach, this is an artefact of the Fourier space, this will become more evident in the course of this section and in the next where we deal with the characteristics in the real phase space.
Different from the previous case however, we notice that $F \neq 0$:

$$F = \frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'l}{k'^2}\left[f_{0,com}\delta f^{(1)\Phi} + f^{\Phi}_{0,com}\delta f^{(1)}\right] \tag{4.14}$$

we notice that F is a linear operator in $\delta f^{(1)}$ and since we do not have a source term, this is completely homogenous, which lets us write a solution of the form:

$$\delta f^{(1)} = \tau \exp(\int_0^\eta F d\eta')\delta f^{(1)}(\eta_0) := U(\eta)\delta f^{(1)}(\eta_0) \tag{4.15}$$

where it should be stated that we use the series expansion for the exponential function, since F is an operator which acts on $\delta f^{(1)}$

4.1.3 Zel'dovich perturbed trajectories

In this section we will attempt to find a solution for the full solution of the Vlasov equation under a perturbed Zel'dovich split. We start by writing down the full Vlasov equation again:

$$\mathcal{L}_0 f + \mathcal{L}_{Grav} = 0 \tag{4.16}$$

$$\Leftrightarrow \quad \mathcal{L}_0 \left[f_{0,Z} + \delta_Z f_Z\right] + \mathcal{L}_{Grav}\left[f_{0,Z} + \delta_Z f_Z\right] = 0 \tag{4.17}$$

$$\Leftrightarrow \quad \mathcal{L}_0 f_{0,Z} + \mathcal{L}^{(1)}_{Grav} f_{0,Z} + \mathcal{L}_0 \delta_Z f_Z + \mathcal{L}_{Grav}\delta_Z f_Z + \sum_{n\geq 2} f_{0,Z} = 0 \tag{4.18}$$

where the first two terms drop out as seen in (3.3), leaving us with:

$$\mathcal{L}_0 \delta_Z f_Z = -\left[\mathcal{L}_{Grav}\delta_Z f_Z + \sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav} f_{0,Z}\right] \tag{4.19}$$

$$\Leftrightarrow \left[\frac{\partial}{\partial \eta} - k\cdot\nabla_l + \frac{1}{2}l\cdot\nabla_l\right]\delta_Z f_Z = \frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'l}{k'^2}\left[f^\Phi \delta_Z f_Z\right] - \sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav} f_{0,Z} = 0 \tag{4.20}$$

, where $\sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav} f_{0,Z}$ can be expressed as:

$$\mathcal{L}_{Grav} f_{0,Z} = -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'l}{k'^2}\left[f^\Phi f_{0,Z}\right] \tag{4.21}$$

$$= -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'l}{k'^2}\left[(f^\Phi_{0,Z} + \delta_Z f^\Phi_Z)(f_{0,com} + \delta f^{(1)})\right] \tag{4.22}$$

$$= -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'l}{k'^2}\left[(f^\Phi_{0,com} + \delta f^{(1)\Phi} + \delta_Z f^\Phi_Z)(f_{0,com} + \delta f^{(1)})\right] \tag{4.23}$$

$$\tag{4.24}$$

from which it follows after factorisation and only taking starting from second order perturbations:

$$\sum_{n\neq 2}\mathcal{L}^{(n)}_{Grav} f_{0,Z} = -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'l}{k'^2}\left[\delta f^{(1)\Phi}\delta f^{(1)} + f_{0,com}\delta_Z f^\Phi_Z + \delta f^{(1)}\delta_Z f^\Phi_Z\right] \tag{4.25}$$

Similarly to the Zel'dovich 'free' streaming, we see that $\delta_Z f_Z$ determines the entirety of the dynamics. We can insert (4.25) into (4.20) to get the full expression of the dynamics which pertains to $\delta_Z f_Z$, for the sake of clarity we will be abbreviating some terms:

$$\mathcal{L}_0 \delta_Z f_Z = \mathcal{L}^H_{Grav}\delta_Z f_Z + \mathcal{L}^S_{Grav}\delta_Z f_Z + I \tag{4.26}$$

Where H denotes the homogenous/linear gravitational operator in $\delta_Z f_Z$, S the gravitational operator which has a squared dependency in $\delta_Z f_Z$ and I which denotes the inhomogeneous gravitational operator aka. the operator which has no dependency on $\delta_Z f_Z$
Depending on how we choose to deal with the squared dependency, where we either ignore it or we don't, we have two different solutions for the Zel'dovich perturbed density function we should get:

$$\frac{d}{d\eta}\delta_Z f_Z(\eta) = M_Z(\eta)\delta_Z f_Z(\eta) + I(\eta) \tag{4.27}$$

such that a solution can be given by:

$$\delta_Z f_Z(\eta) = U_Z(\eta)\delta_Z f_Z + \int_0^\eta d\eta' U_Z(\eta')I(\eta) \tag{4.28}$$

where the operator $U(\eta)$ is given by:

$$U_Z(\eta) = \tau \exp(\int_0^\eta \mathcal{L}^H_{Grav}) \tag{4.29}$$

in the case that we chose to ignore the squared contribution, however if we choose to include this squared contribution we need to take a look at Riccati-type ODE's. For now we leave this section as is and return to it if needed.

4.2 Standard phase space characteristics

We can also choose to apply the method of characteristics in standard phase space, which will help us more for the upcoming section of this thesis, not to say that the method of characteristics is in any way redundant. Similar to the previous section we will work ourselves up to the final solution:

Search for trajectories $(\eta(s), \mathbf{q}(s), \mathbf{p}(s))$ such that

$$\frac{df}{ds}(\eta, \mathbf{q}, \mathbf{p}) = F(f, \eta, \mathbf{q}, \mathbf{p}) \tag{4.1}$$

$$\implies \frac{df}{ds} = \frac{\partial f}{\partial \eta}\frac{d\eta}{ds} + \frac{\partial f}{\partial \mathbf{q}}\frac{d\mathbf{q}}{ds} + \frac{\partial f}{\partial \mathbf{p}}\frac{d\mathbf{p}}{ds} = F \tag{4.2}$$

$$\boxed{\frac{d\eta}{ds} = 1; \quad \frac{d\mathbf{q}}{ds} = \frac{\mathbf{p}}{m}; \quad \frac{d\mathbf{p}}{ds} = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\tilde{\Phi}}} \tag{4.3}$$

$\implies F = 0 \implies \frac{df}{d\eta} = 0 \implies f(\eta, \mathbf{q}, \mathbf{p}) = f_0(\eta_0, \mathbf{q}_0, \mathbf{p}_0)$

4.2.1 Free Streaming

Set $\tilde{\Phi} = 0$ and thus $\nabla_{\mathbf{q}}\tilde{\Phi} = 0$:

$$\boxed{\frac{d\eta}{ds} = 1; \quad \frac{d\mathbf{q}}{ds} = \frac{\mathbf{p}}{m}; \quad \frac{d\mathbf{p}}{ds} = -\frac{\mathbf{p}}{2}; \quad \frac{df}{ds} = 0} \tag{4.4}$$

$\Longrightarrow$ $s = \eta$ if $\eta(0) = 0$.

$$\mathbf{p}(\eta) = \mathbf{p}(\eta_0)\, e^{-\eta/2} = \mathbf{p}_0\, e^{-\eta/2} \tag{4.5}$$

$$\mathbf{q}(\eta) = \mathbf{q}_0 + \int_0^\eta \frac{\mathbf{p}(\eta')}{m}\, d\eta' = \mathbf{q}_0 + \frac{\mathbf{p}_0}{m} \int_0^\eta e^{-\eta'/2}\, d\eta' = \mathbf{q}_0 + 2\frac{\mathbf{p}_0}{m}\left[1 - e^{-\eta/2}\right] \tag{4.6}$$

$\Longrightarrow$ In free streaming we read off the trajectories:

$$\left(\eta,\ \mathbf{q}_0 + 2\frac{\mathbf{p}_0}{m}\left[1 - e^{-\eta/2}\right],\ \mathbf{p}_0\, e^{-\eta/2}\right) \tag{4.7}$$

along which $f(\eta, \mathbf{q}(\eta), \mathbf{p}(\eta)) = f(0, \mathbf{q}_0, \mathbf{p}_0)$.

CHAPTER 5

Deviations from Zel'dovich Trajectories

Our next step is to get an expression for how the distribution function changes along the characteristics if we perturb the particle trajectories (characteristics).
Our final expression will look like

$$\begin{aligned} f(\mathbf{q},\mathbf{p},t) &= f_{ini}\Big(\mathbf{q}\big(\mathbf{q}^{(i)},\mathbf{p}^{(i)},t\big),\ \mathbf{p}\big(\mathbf{q}^{(i)},\mathbf{p}^{(i)},t\big)\Big) \\ &= f_{ini}\Big(\mathbf{q}_Z+\delta\mathbf{q},\ \mathbf{p}^{(i)}+\delta\mathbf{p}\Big) \\ &\approx f_{ini}\Big(\mathbf{q}_Z,\mathbf{p}^{(i)}\Big)+\partial_{\mathbf{q}}f_{ini}\cdot\delta\mathbf{q}+\partial_{\mathbf{p}}f_{ini}\cdot\delta\mathbf{p} \end{aligned} \tag{5.1}$$

where

$$\mathbf{q}=\mathbf{q}^{(i)}+\mathbf{D}+\mathbf{p}^{(i)}+\delta\mathbf{q}=\mathbf{q}_Z+\delta\mathbf{q} \tag{5.2}$$

$$\mathbf{p}=\mathbf{p}^{(i)}+\delta\mathbf{p} \qquad\qquad =\mathbf{p}^{(i)}+\delta\mathbf{p} \tag{5.3}$$

So for our purposes we need to find expressions for $\delta\mathbf{q}$ and $\delta\mathbf{p}$, since we already know f_0.
We will be using the EdS approximation of the Vlasov equation under new time $\eta=\log\Big(\frac{D(\tau)}{D^{(i)}}\Big)$, where τ is the conformal time.

$$\boxed{\frac{df}{d\eta}=\frac{\partial f}{\partial\eta}+\frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{q}}f-\left[\frac{\mathbf{p}}{2}+m\,\nabla_{\mathbf{q}}\tilde{\Phi}\right]\cdot\nabla_{\mathbf{p}}f=0} \tag{5.4}$$

5.1 Towards the Zel'dovich Trajectories

REMINDER: We want to get to the Zel'dovich trajectories.
On our way there perhaps it would be wise to start from the standard comoving split: $f=f_0+\delta f$.
Note: for now we will be dealing with **LINEAR THEORY**.
We will continue by defining f_0 by its symmetries:

- homogeneity in space
- isotropy in momentum
- solves Vlasov eq. without gravity

$$\frac{df_0}{d\eta} = 0 \implies \frac{\partial f_0}{\partial \eta} + \underbrace{\frac{\mathbf{p}}{m} \cdot \nabla_{\mathbf{q}} f_0}_{=0} - \frac{\mathbf{p}}{2} \cdot \nabla_{\mathbf{p}} f_0 = 0 \tag{5.1}$$

$$\iff \frac{\partial f_0}{\partial \eta} - \frac{\mathbf{p}}{2} \cdot \nabla_{\mathbf{p}} f_0 = 0 \quad \longrightarrow \quad \text{specific solution of free streaming} \tag{5.2}$$

$$\implies \frac{df}{d\eta} = \frac{d}{d\eta}[f_0 + \delta f] = 0 \iff \frac{\partial}{\partial \eta}[f_0 + \delta f] + \frac{\mathbf{p}}{m} \nabla_{\mathbf{q}}[f_0 + \delta f] - \left[\frac{\mathbf{p}}{2} + m\, \nabla_{\mathbf{q}} \tilde{\Phi}\right] \nabla_{\mathbf{p}}[f_0 + \delta f] = 0 \tag{5.3}$$

$$\iff \underbrace{\frac{\partial}{\partial \eta} f_0 - \frac{\mathbf{p}}{2} \nabla_{\mathbf{p}} f_0}_{=0} + \frac{\partial}{\partial \eta} \delta f + \frac{\mathbf{p}}{m} \nabla_{\mathbf{q}} \delta f - \left[\frac{\mathbf{p}}{2} + m \nabla_{\mathbf{q}} \tilde{\Phi}\right] \nabla_{\mathbf{p}} \delta f - m\, \nabla_{\mathbf{q}} \tilde{\Phi}\, \nabla_{\mathbf{p}} f_0 = 0 \tag{5.4}$$

$$\iff \frac{\partial}{\partial \eta} \delta f + \frac{\mathbf{p}}{m} \nabla_{\mathbf{q}} \delta f - \left[\frac{\mathbf{p}}{2} + m\, \nabla_{\mathbf{q}} \tilde{\Phi}\right] \nabla_{\mathbf{p}} \delta f - m\, \nabla_{\mathbf{q}} \tilde{\Phi}\, \nabla_{\mathbf{p}} f_0 = 0 \tag{5.5}$$

$$\iff \frac{d}{d\eta}[\delta f] = m\, \nabla_{\mathbf{q}} \tilde{\Phi}\, \nabla_{\mathbf{p}} f_0 \tag{5.6}$$

It could be wise to also split $\tilde{\Phi}$ into background + perturbation, since it is linear in f.
As we know

$$\Delta \tilde{\Phi} = 4\pi G \bar{\rho}\, a^2 \delta = 4\pi G a^2 (\rho - \bar{\rho}) \tag{5.7}$$

where $\rho = m \int d^3p\, f(\mathbf{q}, \mathbf{p}, \eta)$ and $\tilde{\Phi} = \Phi/(CHf_+)^2$ and $\bar{\rho} = m \int d^3p f_0$ by definition.

$$\begin{aligned}
\implies \Delta \tilde{\Phi} &= \frac{4\pi G a^2}{(Hf_+)^2} (\rho - \bar{\rho}) && (5.8)\\
&= c\, (\rho - \bar{\rho}) && (5.9)\\
&= c \left(m \int d^3p\, [f_0 + \delta f] - \bar{\rho} \right) && (5.10)\\
&= c \left(m \int d^3p\, f_0 + m \int d^3p\, \delta f - \bar{\rho} \right) \\
&= c\, \delta \rho && (5.11)
\end{aligned}$$

$$\implies \delta \tilde{\Phi} = c \int d^3q'\, G(\mathbf{q} - \mathbf{q}')\, \delta \rho(\mathbf{q}', \eta), \quad \text{where} \quad G(\mathbf{q} - \mathbf{q}') = -\frac{1}{4\pi |\mathbf{q} - \mathbf{q}'|} \tag{5.12}$$

$$\begin{aligned}
\implies \nabla_{\mathbf{q}} \tilde{\Phi} &= c \int d^3q'\, \nabla_{\mathbf{q}} G(\mathbf{q} - \mathbf{q}')\, \delta \rho(\mathbf{q}', \eta) \\
&= \frac{c}{4\pi} \int d^3q'\, \frac{\mathbf{q} - \mathbf{q}'}{|\mathbf{q} - \mathbf{q}'|^3} \int d^3p\, \delta f(\mathbf{q}', \mathbf{p}, \eta) \\
&= \mathcal{C} \iint d^3q'\, d^3p\, \frac{\mathbf{q} - \mathbf{q}'}{|\mathbf{q} - \mathbf{q}'|^3}\, \delta f(\mathbf{q}', \mathbf{p}, \eta) && (5.13)
\end{aligned}$$

$$\equiv \nabla_{\mathbf{q}}\tilde{\Phi}[\delta f] \equiv \nabla_{\mathbf{q}}\delta\tilde{\Phi} \tag{5.14}$$

Note: linear in δf but independent of $\mathbf{p}$.
Results in the same thing, since $\nabla_{\mathbf{q}}\bar{\rho} = 0$ and $\nabla_{\mathbf{q}} f_0 = 0 \implies$ spatial homogeneity.
We notice that the gravitational potential is completely independent of f_0 and is entirely given by the perturbation δf.

$$\implies \frac{\partial}{\partial\eta}\delta f + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}}\delta f - \left[\frac{\mathbf{p}}{2} + m\,\nabla_{\mathbf{q}}\delta\tilde{\Phi}\right]\nabla_{\mathbf{p}}\delta f - m\,\nabla_{\mathbf{q}}\delta\tilde{\Phi}\,\nabla_{\mathbf{p}} f_0 = 0 \tag{5.15}$$

In linear theory we drop all terms squared in perturbation (first order consideration only):

$$\implies \frac{\partial}{\partial\eta}\delta f + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}}\delta f - \frac{\mathbf{p}}{2}\nabla_{\mathbf{p}}\delta f - m\,\nabla_{\mathbf{q}}\delta\tilde{\Phi}\,\nabla_{\mathbf{p}} f_0 = 0 \tag{5.16}$$

We get the characteristics given by

$$\frac{d\eta}{ds} = 1; \quad \frac{d\mathbf{q}}{d\eta} = \frac{\mathbf{p}}{m}; \quad \frac{d\mathbf{p}}{d\eta} = -\frac{\mathbf{p}}{2} \tag{5.17}$$

$$\implies s = \eta; \quad \mathbf{q}(\eta) = \mathbf{q}_0 + 2\frac{\mathbf{p}_0}{m}\left[1 - e^{-\eta/2}\right]; \quad \mathbf{p}(\eta) = \mathbf{p}_0\, e^{-\eta/2} \tag{5.18}$$

but notably we get

$$\frac{d}{d\eta}\delta f = F = m\,\nabla_{\mathbf{q}}\delta\tilde{\Phi}\,\nabla_{\mathbf{p}} f_0 \tag{5.19}$$

We can choose $F = B[\delta f]$, where $B = m\,\nabla_{\mathbf{p}} f_0 \cdot \mathcal{C}\iint d^3q'\, d^3p\, \dfrac{\mathbf{q}-\mathbf{q}'}{|\mathbf{q}-\mathbf{q}'|^3}(\cdot)$,
which is a linear operator in δf such that we can write a solution of the form

$$\boxed{\delta f = \mathcal{T}\exp\left(\int_{\eta_0}^{\eta} B\, d\eta'\right)\delta f(\eta_0)} \tag{5.20}$$

or

$$\delta f = U(\eta, \eta_0)\delta f_{ini} \tag{5.21}$$

along the previously mentioned characteristics, where the exponential is expressed as a series expansion. We also reffer to $\delta f(\eta_0)$ as δf_{ini}

5.2 Zel'dovich "Free Streaming"

In this approach we look at the case where we say that the first order gravitational contribution is included into the background, so once again we have to come up with some kind of idea for a background.
We first look at the general "free-streaming" case and see whether we can derive some notion of a background from it.
Vlasov:

$$\frac{\partial f}{\partial\eta} + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}} f - \left[\frac{\mathbf{p}}{2} + m\,\nabla_{\mathbf{q}}\tilde{\Phi}\right]\nabla_{\mathbf{p}} f = 0 \tag{5.1}$$

As I see it there is only one approach to this: we have to look at our analysis in Fourier space and just take our approach from there, where we stated:

$$f_Z = f_0 + \delta f^{(1)}; \quad \delta f_Z = \delta f^{(2)} + \delta f^{(3)} + \cdots \tag{5.2}$$

and f_Z is the function which solves

$$\frac{\partial f_Z}{\partial \eta} + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}} f_Z - \left[\frac{\mathbf{p}}{2} + m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}\right]\nabla_{\mathbf{p}} f_Z = 0 \tag{5.3}$$

$$\begin{aligned} \Longrightarrow \frac{\partial}{\partial \eta} f_0 + \frac{\partial}{\partial \eta}\delta f^{(1)} + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}} f_0 + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}}\delta f^{(1)} - \frac{\mathbf{p}}{2}\nabla_{\mathbf{p}}\delta f^{(1)} \\ - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}\nabla_{\mathbf{p}} f_0 - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}\nabla_{\mathbf{p}}\delta f^{(1)} = 0 \end{aligned} \tag{5.4}$$

$$\Longleftrightarrow \frac{\partial}{\partial \eta}\delta f^{(1)} + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}}\delta f^{(1)} - \left[\frac{\mathbf{p}}{2} + m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}\right]\nabla_{\mathbf{p}}\delta f^{(1)} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}\,\nabla_{\mathbf{p}} f_0 = 0 \tag{5.5}$$

$$\Longleftrightarrow \frac{\partial}{\partial \eta}\delta f^{(1)} + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}}\delta f^{(1)} - \frac{\mathbf{p}}{2}\nabla_{\mathbf{p}}\delta f^{(1)} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}\,\nabla_{\mathbf{p}} f_0 = 0 \tag{5.6}$$

→ We wipe out $\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}\,\nabla_{\mathbf{p}}\delta f^{(1)}$ in linear theory.
→ We notice that this is **completely analogous** to the previous case, except with $\tilde{\Phi}^{(1)}$ instead of $\tilde{\Phi}$.
→ We are kind-of stuck here, because we need a full expansion of the Vlasov eq. → we cannot stop at the boundary of the linear theory, since it will not yield the Zel'dovich trajectories from the characteristics.
→ **We take a step back!** → **Give up linear theory.**

$$\frac{\partial f_Z}{\partial \eta} + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}} f_Z - \left[\frac{\mathbf{p}}{2} + m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}\right]\nabla_{\mathbf{p}} f_Z = 0 \tag{5.7}$$

→ Method of characteristics:

$$\boxed{\frac{d\eta}{ds} = 1; \quad \frac{d\mathbf{q}}{ds} = \frac{\mathbf{p}}{m}; \quad \frac{d\mathbf{p}}{ds} = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)} \quad \text{and} \quad \frac{df}{d\eta} = 0} \tag{5.8}$$

$$\Longrightarrow f(\eta, \mathbf{p}_Z(\eta), \mathbf{q}_Z(\eta)) = f(\eta_0, \mathbf{p}_0, \mathbf{q}_0) \quad \longrightarrow \quad \text{distr. conserved along char.} \tag{5.9}$$

The trajectories are given from the set of differential eq.

CHAPTER 6

Perturbed Paths

Goal: On the basis of the trajectories $\mathbf{q}_Z$ and $\mathbf{p}_Z$ we wish to evaluate the evolution of f along a path slightly perturbed from these.
We introduce: $\mathbf{q}(\eta) = \mathbf{q}_Z(\eta) + \delta\mathbf{q}(\eta)$; $\mathbf{p}(\eta) = \mathbf{p}_Z(\eta) + \delta\mathbf{p}(\eta)$.

$$\begin{aligned}
\implies \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}) &= \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z + \delta\mathbf{q}) \\
&\approx \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z) + \nabla_{\mathbf{q}}\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q} \\
&= \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z) + \frac{\partial^2\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z)}{\partial_i\partial_j}\,\delta\mathbf{q} \qquad (6.1) \\
&= \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z) + \mathbf{H}[\tilde{\Phi}^{(1)}]\delta\mathbf{q} \qquad (6.2)
\end{aligned}$$

Where $H[\tilde{\Phi}^{(1)}]$ denotes the Hessian of $\tilde{\Phi}$. From now on we will be referring to this Hessian as $H^{(1)}$ which naturally retains its time and positional dependence.

$$\text{note:} \quad \frac{d\mathbf{q}}{d\eta} = \frac{\mathbf{p}}{m} \implies \frac{d}{d\eta}\delta\mathbf{q} = \frac{\delta\mathbf{p}}{m} \qquad (6.3)$$

and

$$\frac{d\mathbf{p}}{d\eta} = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}) \qquad (6.4)$$

$$\begin{aligned}
\implies \frac{d}{d\eta}\mathbf{p}_Z + \frac{d}{d\eta}\delta\mathbf{p} &= -\frac{\mathbf{p}_Z}{m} - \frac{\delta\mathbf{p}}{m} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z) - m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q} \\
&\approx -\left[\frac{\mathbf{p}_Z}{m} + \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z)\cdot m\right] - \left[\frac{\delta\mathbf{p}}{m} + m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}\right] \qquad (6.5)
\end{aligned}$$

$$\boxed{\implies \frac{\partial}{\partial\eta}\delta\mathbf{p} \approx -\frac{\delta\mathbf{p}}{m} - \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}\cdot m} \qquad (6.6)$$

We have: $\frac{d}{d\eta}\delta\mathbf{q} = \frac{\delta\mathbf{p}}{m} \implies \delta\mathbf{q} = \frac{\delta\mathbf{p}}{m}\eta + \delta\mathbf{q}_0$

$$\begin{aligned}
\implies \frac{d}{d\eta}\delta\mathbf{p} &\approx -\frac{\delta\mathbf{p}}{m} - \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\cdot m\left[\frac{\delta\mathbf{p}}{m}\eta + \delta\mathbf{q}_0\right] \\
&= -\frac{\delta\mathbf{p}}{m} - \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{p}\,\eta - m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}_0 \\
&= -\delta\mathbf{p}\left[\frac{1}{m}\mathbf{I} + \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\eta\right] - m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}_0 \qquad (6.7)
\end{aligned}$$

6.1 Variation of Constants

$$\frac{d}{d\eta}\delta\mathbf{p} = -\delta\mathbf{p}\underbrace{\left[\frac{1}{m}\mathbf{I} + \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\eta\right]}_{=-a(\eta)} - \underbrace{m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}_0}_{=b(\eta)} \qquad (6.1)$$

$$\implies \delta\mathbf{p}(\eta) = \delta\mathbf{p}_H(\eta)\Big(\delta\mathbf{p}_0 + \int_0^\eta -m\mathbf{H}^{(1)}(\eta', \mathbf{q}_Z)\,\delta\mathbf{p}_H^{-1}\,\delta\mathbf{q}_0(\eta')\,d\eta'\Big) \qquad (6.2)$$

$$\delta\mathbf{p}_H(\eta) = \exp\Big(\int_0^\eta -\left[\frac{1}{m}\mathbf{I} + \mathbf{H}(\eta', \mathbf{q}_Z)\,\eta'\right]d\eta'\Big) \qquad (6.3)$$

We can show that this is linear in δf, since

$$\begin{aligned}
H^{(1)} &= \frac{\partial^2\tilde{\Phi}^{(1)}}{\partial_i\partial_j} && (6.4)\\
&= c\int d^3q'\,\frac{\partial^2 G(\mathbf{q}, \mathbf{q}')}{\partial_i\partial_j}\,\delta\rho^{(1)}(\mathbf{q}', \eta) && (6.5)\\
&= c\int d^3q'\,\frac{\partial^2 G(\mathbf{q}, \mathbf{q}')}{\partial_i\partial_j}\,m\int d^3p\,\delta f^{(1)}(\mathbf{q}', \eta) && (6.6)\\
&= cm\int\int d^3q'd^3p\,\frac{\partial^2 G(\mathbf{q}, \mathbf{q}')}{\partial_i\partial_j}\,\delta f^{(1)}(\mathbf{q}', \eta) && (6.7)\\
&= cm\int\int d^3q'd^3p\,\frac{\partial^2 G(\mathbf{q}, \mathbf{q}')}{\partial_i\partial_j}\,U(\eta, \eta_0)\delta f^{(1)}_{ini}(\mathbf{q}') && (6.8)\\
&\equiv \mathbf{P}[\delta f^{(1)}_{ini}](\eta) \quad \text{or} \quad = \mathbf{P}(\eta)\delta f^{(1)}_{ini} && (6.9)\\
& && (6.10)
\end{aligned}$$

the **Poisson operator**.

$$\implies \delta\mathbf{p}_H(\eta) = \exp\Big(-\frac{\eta}{m}\mathbf{I}\Big)\exp\Big(-\int_0^\eta P(\eta')\,\delta f^{(1)}_{\text{ini}}\,\eta'\,d\eta'\Big) \qquad (6.11)$$

$$\boxed{\implies \delta\mathbf{p}(\eta) = \delta\mathbf{p}_H(\eta)\Big(\delta\mathbf{p}_0 + \int_0^\eta -m\,P(\eta')\,\delta f^{(1)}_{\text{ini}}\,\delta\mathbf{p}_H^{-1}\delta\mathbf{q}_0\,d\eta'\Big)} \qquad (6.12)$$

$$\boxed{\implies \delta\mathbf{q} = \frac{\delta\mathbf{p}(\eta)}{m}\,\eta + \delta\mathbf{q}_0} \qquad (6.13)$$

These are the last step in determining the evolution of f along the perturbed characteristics. We remind:

$$f(\mathbf{q}, \mathbf{p}, \eta) \approx \underbrace{f_0(\mathbf{q}_Z, \mathbf{p}^{(i)})}_{\substack{\text{Initial distribution} \\ \text{(not background)}}} + \partial_{\mathbf{q}} f_0 \cdot \delta\mathbf{q} + \partial_{\mathbf{p}} f_0 \cdot \delta\mathbf{p} \tag{6.14}$$

6.2 Final Remarks

One might notice that we are taking the propagator $U(\eta)$ generated from the linear theory and one might ask why, since we stated that we have departed from linear theory and if said propagator generates the correct $\delta f^{(1)}(\eta)$ for our case.
The answer is yes. Since we are dealing with the Zel'dovich trajectories generated from the Zel'dovich "free-streaming" ansatz for the Vlasov equation, we by definition stop at the first order gravitational perturbation $\delta f^{(1)}$ and disregard all higher order perturbations such that this propagator is the correct choice.
In spite of the correct formulation with respect to the Zel'dovich frames one might notice that we have restricted ourselves to the first order gravitational contributions, such that even if we are in fact analysing paths deviating from the Zel'dovich trajectories we can only ever hope to analyse dynamics which arise from the first order Hessian $\mathbf{H}^{(1)}$. If we want to take a look at what dynamics might be produced from higher order gravitational contributions, we must proceed with a Born approximation.

CHAPTER 7

Born Approximation for Perturbed Trajectories

In this approximation we will attempt to get to a tractable expression for the perturbed trajectories. We will compute the effect of a perturbation by evaluating it along the unperturbed solution (Zel'dovich trajectories).
We start off assuming that the change in position is given by a velocity field and that the change in momentum is given by a force field:

$$\dot{\mathbf{q}} = \mathbf{v}; \quad \dot{\mathbf{p}} = \mathbf{F}(\mathbf{q}, \mathbf{p}, \eta) \quad \Longrightarrow \quad \dot{\mathbf{q}} = \frac{\mathbf{p}}{m}; \quad \dot{\mathbf{p}} = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi} \tag{7.1}$$

We get the right hand side from the characteristics of the Einstein–de Sitter Universe, where time is $\eta = \log\left(\frac{D(\tau)}{D^{(i)}}\right)$.

$$\tilde{\Phi} = s\tilde{\Phi}^{(1)} + \varepsilon\,\delta\tilde{\Phi} \quad \Longrightarrow \quad \mathbf{F}_0 = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}, \quad \mathbf{F}_1(\mathbf{q}, \eta) = -m\,\nabla_{\mathbf{q}}\delta\tilde{\Phi} \tag{7.2}$$

$$\Longrightarrow \mathbf{F} = \mathbf{F}_0 + \varepsilon\,\mathbf{F}_1 \tag{7.3}$$

$$\Longrightarrow \dot{\mathbf{q}}_Z = \frac{\mathbf{p}_Z}{m}; \quad \dot{\mathbf{p}}_Z = \mathbf{F}_0(\mathbf{q}_Z, \mathbf{p}_Z, \eta) \tag{7.4}$$

$$\mathbf{q} = \mathbf{q}_Z + \varepsilon\,\delta\mathbf{q} \qquad \mathbf{p} = \mathbf{p}_Z + \varepsilon\,\delta\mathbf{p} \tag{7.5}$$

$$\begin{aligned}\dot{\mathbf{p}} = \dot{\mathbf{p}}_Z + \varepsilon\,\delta\dot{\mathbf{p}} &= \mathbf{F}_0(\mathbf{q}_Z + \varepsilon\,\delta\mathbf{q},\ \mathbf{p}_Z + \varepsilon\,\delta\mathbf{p},\ \eta) + \varepsilon\,\mathbf{F}_1(\mathbf{q}_Z + \varepsilon\,\delta\mathbf{q},\ \eta) \\ &= \mathbf{F}_0(\mathbf{q}_Z, \mathbf{p}_Z) + \varepsilon\left[\left.\frac{\partial \mathbf{F}_0}{\partial \mathbf{q}}\right|_Z \delta\mathbf{q} + \left.\frac{\partial \mathbf{F}_0}{\partial \mathbf{p}}\right|_Z \delta\mathbf{p}\right] + \varepsilon\,\mathbf{F}_1(\mathbf{q}_Z, \eta) + \mathcal{O}(\varepsilon^2) \\ &= -\frac{\mathbf{p}_Z}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)} + \varepsilon\left[-m \underbrace{\frac{\partial^2 \tilde{\Phi}^{(1)}}{\partial q_i\,\partial q_j}}_{=H[\tilde{\Phi}^{(1)}]=H^{(1)}} \delta\mathbf{q} - \frac{1}{2}\delta\mathbf{p} + \mathbf{F}_1(\mathbf{q}_Z, \eta)\right] + \mathcal{O}(\varepsilon^2)\end{aligned} \tag{7.6}$$

$$\implies \delta\dot{\mathbf{p}} = -m\,H^{(1)}\delta\mathbf{q} - \frac{1}{2}\delta\mathbf{p} + \mathbf{F}_1(\mathbf{q}_Z, \eta); \qquad \delta\dot{\mathbf{q}} = \frac{\delta\mathbf{p}}{m} \tag{7.7}$$

$$\boxed{\frac{d}{d\eta}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} = \underbrace{\begin{pmatrix}0 & \frac{1}{m}\mathbf{I}\\ -m\,H^{(1)} & -\frac{1}{2}\mathbf{I}\end{pmatrix}}_{\equiv M(\eta)}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} + \begin{pmatrix}0\\ \mathbf{F}_1(\mathbf{q}_Z, \eta)\end{pmatrix}} \tag{7.8}$$

This differential equation gets solved by a Volterra type integral:

$$\boxed{\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} = \Psi(\eta, \eta_0)\begin{pmatrix}\delta\mathbf{q}_0\\ \delta\mathbf{p}_0\end{pmatrix} + \int_{\eta_0}^{\eta}\Psi(\eta, s)\begin{pmatrix}0\\ \mathbf{F}_1(\mathbf{q}_Z(s), s)\end{pmatrix} ds} \tag{7.9}$$

$$\approx \int_{\eta_0}^{\eta}\Psi(\eta, s)\begin{pmatrix}0\\ \mathbf{F}_1(\mathbf{q}_Z(s), s)\end{pmatrix} ds \quad \longleftarrow \quad \text{we assume no initial perturbations} \tag{7.10}$$

where $\Psi(\eta, \eta_0)$ is the propagator which we define as:

$$\partial_\eta\,\Psi(\eta, \eta') = M(\eta')\,\Psi(\eta, \eta'); \qquad \Psi(\eta', \eta') = \mathbf{I} \tag{7.11}$$

$$\implies \Psi(\eta, \eta') = \exp\left(\int_{\eta'}^{\eta} M(\eta^*)\,d\eta^*\right) \quad \longleftarrow \quad \text{series expansion} \tag{7.12}$$

The next step is to find an appropriate truncation of this series expansion. We drop any contributions which are squared in the Hessian $H^{(1)}$ (or higher), since the Hessian depends component-wise on $\delta f^{(1)}$.

7.1 Decomposition of the Propagator

To proceed we must decompose $M(\eta)$:

$$M(\eta) = \begin{pmatrix}0 & \frac{1}{m}\mathbf{I}\\ -m\,H^{(1)} & -\frac{1}{2}\mathbf{I}\end{pmatrix} = \underbrace{\begin{pmatrix}0 & \frac{1}{m}\mathbf{I}\\ 0 & -\frac{1}{2}\mathbf{I}\end{pmatrix}}_{\equiv M_0} + \underbrace{\begin{pmatrix}0 & 0\\ -m\,H^{(1)} & 0\end{pmatrix}}_{\equiv M_1(\eta)} \tag{7.1}$$

7.1.1 The Homogeneous Solution Ψ

$$\frac{d}{d\eta}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} = M(\eta)\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} \quad \Longleftrightarrow \quad \frac{d}{d\eta}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} = (M_0 + M_1)\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} \tag{7.2}$$

By looking we find the free evolution propagator:

$$\Psi_0 = \exp\Big(M_0(\eta - \eta_0)\Big) \tag{7.3}$$

What about the rest of the evolution? (i.e. non-free evo.)

$$\begin{pmatrix}\delta\tilde{\mathbf{q}}\\ \delta\tilde{\mathbf{p}}\end{pmatrix} = \Psi_0^{-1}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} = e^{-M_0(\eta-\eta_0)}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} \tag{7.4}$$

$$\begin{aligned}
\Longrightarrow \frac{d}{d\eta}\begin{pmatrix}\delta\tilde{\mathbf{q}}\\ \delta\tilde{\mathbf{p}}\end{pmatrix} &= \frac{d}{d\eta}\left[\Psi_0^{-1}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix}\right] = \left[\frac{d}{d\eta}\Psi_0^{-1}\right]\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} + \Psi_0^{-1}\frac{d}{d\eta}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix}\\
&= -M_0\,\Psi_0^{-1}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} + \Psi_0^{-1}(M_0 + M_1)\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix}\\
&= \Psi_0^{-1}\,M_1\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix}\\
&= \Psi_0^{-1}\,M_1\,\Psi_0\begin{pmatrix}\delta\tilde{\mathbf{q}}\\ \delta\tilde{\mathbf{p}}\end{pmatrix}
\end{aligned} \tag{7.5}$$

$$\Longrightarrow \frac{d}{d\eta}\begin{pmatrix}\delta\tilde{\mathbf{q}}\\ \delta\tilde{\mathbf{p}}\end{pmatrix} = \Psi_0^{-1}\,M_1(\eta)\,\Psi_0\begin{pmatrix}\delta\tilde{\mathbf{q}}\\ \delta\tilde{\mathbf{p}}\end{pmatrix} \tag{7.6}$$

7.2 The Non-Free Propagator

The "non-free" propagator follows as:

$$\boxed{\Psi_1 = \exp\left(\int_{\eta_0}^{\eta}\Psi_0^{-1}\,M_1(\eta')\,\Psi_0\,d\eta'\right)} \tag{7.1}$$

$$\Longrightarrow \Psi(\eta,\eta_0) = \Psi_0(\eta,\eta_0)\,\Psi_1(\eta,\eta_0) \qquad \text{let } \tau = \eta - \eta_0 \tag{7.2}$$

$$= e^{M_0\tau}\exp\left(\int_{\eta_0}^{\eta}\underbrace{e^{-M_0\tau(\eta')}\,M_1(\eta')\,e^{M_0\tau(\eta')}}_{\Psi_0^{-1}\,M_1(\eta')\,\Psi_0}\,d\eta'\right) \tag{7.3}$$

We continue by searching for a truncation of this operator, as the argument of the second exponent depends on the Hessian.
We first take a look at $e^{M_0\tau}$ and $e^{-M_0\tau}$:

$$e^{M_0\tau} = \sum_{n=0}^{\infty}\frac{(M_0\tau)^n}{n!} = \mathbf{I} + \sum_{n=1}^{\infty}\frac{M_0^n\,\tau^n}{n!} \tag{7.4}$$

where it can be shown that:

$$M_0^n = \begin{pmatrix} 0 & \frac{1}{(-2)^{n-1}}\frac{1}{m}\mathbf{I} \\ 0 & \frac{1}{(-2)^n}\mathbf{I} \end{pmatrix} \tag{7.5}$$

Top right component:

$$\begin{aligned} \frac{1}{m}\sum_{n=1}^{\infty}\frac{1}{(-2)^{n-1}}\cdot\frac{1}{n!}\,\tau^n &= \frac{1}{m}\sum_{n=1}^{\infty}\frac{-2}{(-2)^n}\cdot\frac{1}{n!}\,\tau^n \\ &= -\frac{2}{m}\sum_{n=1}^{\infty}\left(-\frac{\tau}{2}\right)^n\cdot\frac{1}{n!} \\ &= -\frac{2}{m}\left(e^{-\tau/2}-1\right) \end{aligned} \tag{7.6}$$

Bottom right component:

$$\sum_{n=1}^{\infty}\frac{\tau^n}{(-2)^n}\cdot\frac{1}{n!} = e^{-\tau/2}-1 \tag{7.7}$$

$$\implies \Psi_0 = e^{M_0\tau} = \begin{pmatrix} \mathbf{I} & \frac{2}{m}(1-e^{-\tau/2})\,\mathbf{I} \\ 0 & e^{-\tau/2}\,\mathbf{I} \end{pmatrix} \tag{7.8}$$

$$\implies \Psi_0^{-1} = e^{-M_0\tau} = \begin{pmatrix} \mathbf{I} & \frac{2}{m}(1-e^{\tau/2})\,\mathbf{I} \\ 0 & e^{\tau/2}\,\mathbf{I} \end{pmatrix} \tag{7.9}$$

$$\begin{aligned} \implies \tilde{M}_1(\eta) &= \Psi_0^{-1}\,M_1\,\Psi_0 \\ &= \begin{pmatrix} \mathbf{I} & \frac{2}{m}(1-e^{\tau/2})\,\mathbf{I} \\ 0 & e^{\tau/2}\,\mathbf{I} \end{pmatrix}\begin{pmatrix} 0 & 0 \\ -m\,H^{(1)} & 0 \end{pmatrix}\begin{pmatrix} \mathbf{I} & \frac{2}{m}(1-e^{-\tau/2})\,\mathbf{I} \\ 0 & e^{-\tau/2}\,\mathbf{I} \end{pmatrix} \\ &= \begin{pmatrix} 2(e^{\tau/2}-1)\,H^{(1)} & -\frac{4}{m}(1-e^{\tau/2})(1-e^{-\tau/2})\,H^{(1)} \\ m\,e^{\tau/2}\,H^{(1)} & 2\,e^{\tau/2}(1-e^{-\tau/2})\,H^{(1)} \end{pmatrix} \end{aligned} \tag{7.10}$$

We notice that every component in the block form depends on the Hessian and thus on $\delta f^{(1)}$.
$\implies$ We can truncate at second order.
$\implies$ **We cannot!!!** Later we use $\mathbf{F}_1$, which includes higher orders $\implies$ we are not allowed to truncate.

$$\begin{aligned} \implies \Psi(\eta,\eta_0) &= \Psi_0(\eta,\eta_0)\,\Psi_1(\eta,\eta_0) \\ &= e^{M_0\tau}\exp\!\left(\int_{\eta_0}^{\eta}\Psi_0^{-1}\,M_1\,\Psi_0\,d\eta'\right) \\ &= e^{M_0\tau}\exp\!\left(\int_{\eta_0}^{\eta}\tilde{M}_1(\eta')\,d\eta'\right) \\ &\approx e^{M_0\tau}\left\{\mathbf{I}+\int_{\eta_0}^{\eta}\tilde{M}_1(\eta')\,d\eta'\right\} + \mathcal{O}\!\left((H^{(1)})^2\right) \end{aligned} \tag{7.11}$$

$$\boxed{\implies \Psi(\eta,\eta_0) \approx e^{M_0\tau}\left\{\mathbf{I}+\int_{\eta_0}^{\eta}\tilde{M}_1(\eta')\,d\eta'\right\}} \tag{7.12}$$

7.3 Resolving the Nested Integral

While we have found a valid truncation we are still left with a reappearing problem: our dependence of $H^{(1)}$ on $\delta f^{(1)}$ and thus on the propagator of $\delta f_0^{(1)}$; we once again run into a non-tractable double integral / nested integral.
We can circumvent this by using results/features from hydrodynamics and the Einstein–de Sitter ansatz:

$$\eta = \log\left(\frac{D(\tau)}{D^{(i)}}\right) \quad \Longrightarrow \quad e^{\eta} = \frac{D(\tau)}{D^{(i)}} \tag{7.1}$$

$$\Longrightarrow \; D(\tau) = e^{\eta}\, D^{(i)} \equiv D(\eta) \tag{7.2}$$

It should be noted that τ is the conformal time in this case.
From Cosmological Structure Formation we know:

$$\boxed{\delta = \delta_0\, D_+(a) = \delta_0\, D^{(i)}\, e^{\eta} = \delta_{\text{ini}}\, e^{\eta}} \tag{7.3}$$

In this ansatz we can dodge $\delta f^{(1)}$ and instead make use of $\delta^{(1)}$ $\Longrightarrow$ no more problematic propagator.
Next step is expressing $\tilde{\Phi}^{(1)}$ in terms of $\delta^{(1)} = \delta^{(1)}_{\text{ini}}\, e^{\eta}$:

$$\Delta\tilde{\Phi}^{(1)} = \frac{4\pi G\,\bar{\rho}\,a^2}{(Hf_+)^2}\,\delta^{(1)} \qquad G(\mathbf{q},\mathbf{q}') = -\frac{1}{4\pi|\mathbf{q}-\mathbf{q}'|} \tag{7.4}$$

$$\begin{aligned}
\Longrightarrow \; \tilde{\Phi}^{(1)} &= \frac{4\pi G\,\bar{\rho}\,a^2}{(Hf_+)^2}\int G(\mathbf{q},\mathbf{q}')\,\delta^{(1)}(\mathbf{q}',\eta)\,d^3q' \\
&\equiv c\int G(\mathbf{q},\mathbf{q}')\,\delta^{(1)}(\mathbf{q}',\eta)\,d^3q' \\
&= c\int G(\mathbf{q},\mathbf{q}')\,e^{\eta}\,\delta^{(1)}_{\text{ini}}(\mathbf{q}',\eta_0)\,d^3q' \\
&= e^{\eta}\,c\int G(\mathbf{q},\mathbf{q}')\,\delta^{(1)}_{\text{ini}}(\mathbf{q}',\eta_0)\,d^3q'
\end{aligned} \tag{7.5}$$

$$\Longrightarrow \; H^{(1)}_{ij}(\mathbf{q},\eta) = e^{\eta}\,c\int \frac{\partial^2 G(\mathbf{q},\mathbf{q}')}{\partial q_i\,\partial q_j}\,\delta^{(1)}(\mathbf{q}',\eta_0)\,d^3q' \tag{7.6}$$

$$= e^{\eta}\,H^{(1)}_{\text{ini},\,ij}(\mathbf{q},\eta_0) \tag{7.7}$$

$$\boxed{\Longrightarrow \; H^{(1)}(\mathbf{q},\eta) = e^{\eta}\,H^{(1)}_{\text{ini}}(\mathbf{q},\eta_0)} \tag{7.8}$$

We can now return to our suggested solution:

$$\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} \approx \int_{\eta_0}^{\eta}\Psi(\eta,s)\begin{pmatrix}0\\ \mathbf{F}_1(\mathbf{q}_Z(s),s)\end{pmatrix}ds \tag{7.9}$$

While $\Psi(\eta,s)$ is still an integral, we know it to be tractable, since it depends on known quantities. The only issue left to deal with is $\mathbf{F}_1$, as it depends on $\delta\tilde{\Phi}$, which we can treat analogously to the Hessian.

$$\mathbf{F}_1(\mathbf{q}_Z(s),s) = -m\,\nabla_{\mathbf{q}}\delta\tilde{\Phi}(\mathbf{q},s)\Big|_{\mathbf{q}=\mathbf{q}_Z(s)} \tag{7.10}$$

$$= -m\, e^s \int \nabla_{\mathbf{q}} G(\mathbf{q}, \mathbf{q}')\, \delta_{\text{ini}}^{(n\geq 2)}(\mathbf{q}', \eta_0)\, d^3q' \Big|_{\mathbf{q}=\mathbf{q}_Z(s)} \tag{7.11}$$

which is also tractable.
It should be noted that for the derivation of Ψ we truncate the expression at the second order to retain the Zel'dovich nature of the propagator; yet our final expressions for the perturbed position and momentum depend on F_1 which notably depends on perturbations of order 2 and higher. One might therefore question the validity of the prior truncation, however this procedure retains it validity due to the nature of Born approximations, since we evaluate the effects of perturbations along uperturbed paths.

7.4 Background/Initial Conditions via Gaussian Closure

The final step in finding the approximate solution for f lies in finding an initial condition $f_0^{(i)} = f_0$, which notably remains unchanged along characteristics according to previous calculations. Since our approximate solution $\mathbf{f}$ not only depends on $\mathbf{f_0}$ but also on $\partial_{\mathbf{q}}\mathbf{f_0}$ as well as $\partial_{\mathbf{p}}\mathbf{f_0}$ we must require it to be differentiable in both position and momentum. The SSA solution proves to be unusable due to the non differentiable nature of the Dirac Delta function which appears; however we can make use of the expression that includes cumulants up to the second order. Starting from the general form above and imposing *isotropy*,

$$\sigma_{ij} = \bar{\sigma}^2\, \delta_{ij} \quad \Rightarrow \quad (\sigma^{-1})_{ij} = \frac{1}{\bar{\sigma}^2}\, \delta_{ij}, \tag{7.1}$$

so the distribution simplifies to

$$f_s(\mathbf{q}, \mathbf{p}, \tau) = \frac{n(\mathbf{q}, \tau)}{(2\pi\bar{\sigma}^2)^{3/2}} \exp\left(-\frac{(\mathbf{p} - am\,\mathbf{u})^2}{2\bar{\sigma}^2}\right). \tag{7.2}$$

At the initial time $\eta = 0$ we set

$$n(\mathbf{q}^{(i)}) = \frac{\bar{\rho}}{m}\Big(1 + \delta^{(i)}(\mathbf{q}^{(i)})\Big), \qquad am\,\mathbf{u}^{(i)} = \nabla_q \psi^{(i)} \tag{7.3}$$

where $\psi^{(i)}$ is the initial velocity potential. This gives

$$f_s(\mathbf{q}, \mathbf{p}, \tau) = \frac{\bar{\rho}\Big(1 + \delta^{(i)}(\mathbf{q}^{(i)})\Big)}{(2\pi\bar{\sigma}^2)^{3/2}} \exp\left(-\frac{\Big(\mathbf{p} - \nabla_q \psi^{(i)}\Big)^2}{2\bar{\sigma}^2}\right). \tag{7.4}$$

The density perturbation and the velocity potential are related by

$$\rho^{(i)} = \nabla_q \psi^{(i)}(\mathbf{q}^{(i)}) \;\Rightarrow\; \delta^{(i)} = -\nabla_q^2 \psi^{(i)}, \qquad \Rightarrow\; \psi^{(i)} = -\frac{\phi^{(i)}}{A_\phi}, \quad \text{where } \nabla^2 \phi^{(i)} = A_\phi\, \delta^{(i)}. \tag{7.5}$$

Since $\psi^{(i)}$ is proportional to $\phi^{(i)}$, to maintain first order consistency we need to truncate f_0 at first order in gravitational perturbations. Expanding the Gaussian:

$$\begin{aligned}
\exp\left(-\frac{(\mathbf{p}-\nabla_q\psi^{(i)})^2}{2\bar{\sigma}^2}\right) &= \exp\left(-\frac{p^2 - 2\,\mathbf{p}\cdot\nabla_q\psi^{(i)} + |\nabla_q\psi^{(i)}|^2}{2\bar{\sigma}^2}\right) \\
&\approx \exp\left(-\frac{p^2}{2\bar{\sigma}^2} + \frac{\mathbf{p}\cdot\nabla_q\psi^{(i)}}{\bar{\sigma}^2}\right) \qquad \left(\text{drop } |\nabla_q\psi^{(i)}|^2 \sim \mathcal{O}((\phi^{(i)})^2)\right) \\
&= \exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\exp\left(\frac{\mathbf{p}\cdot\nabla_q\psi^{(i)}}{\bar{\sigma}^2}\right) \\
&\approx \exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(1 + \frac{\mathbf{p}\cdot\nabla_q\psi^{(i)}}{\bar{\sigma}^2}\right), \qquad (\text{exp series truncated at } (\phi^{(i)})^2)
\end{aligned} \tag{7.6}$$

yielding the initial distribution linearised in gravitational perturbations:

$$f_0(\mathbf{q},\mathbf{p},\tau) = \frac{\bar{\rho}\left(1+\delta^{(i)}(\mathbf{q}^{(i)})\right)}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(1 + \frac{\mathbf{p}\cdot\nabla_q\psi^{(i)}}{\bar{\sigma}^2}\right) \tag{7.7}$$

To first order this is

$$\boxed{f_0(\mathbf{q},\mathbf{p},\tau) \approx \frac{\bar{\rho}}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(1 + \delta^{(i)}(\mathbf{q}^{(i)}) + \frac{\mathbf{p}\cdot\nabla_q\psi^{(i)}}{\bar{\sigma}^2}\right)} \tag{7.8}$$

7.4.1 Derivatives of f_0

The spatial and momentum derivatives needed for the Vlasov equation are:

$$\partial_{q_i} f_0 = \frac{\bar{\rho}}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(\partial_{q_i}\delta^{(i)}(\mathbf{q}^{(i)}) + \frac{p_j\,\partial_{q_i}\partial_{q_j}\psi^{(i)}}{\bar{\sigma}^2}\right), \tag{7.9}$$

and, using the product rule on $f_0 = \frac{\bar{\rho}(1+\delta^{(i)})}{(2\pi\bar{\sigma}^2)^{3/2}}e^{-p^2/2\bar{\sigma}^2}(1+\mathbf{p}\cdot\nabla_q\psi^{(i)}/\bar{\sigma}^2)$:

$$\partial_{p_i} f_0 = \frac{\bar{\rho}}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left[\frac{\partial_{q_i}\psi^{(i)}}{\bar{\sigma}^2} - \frac{p_i}{\bar{\sigma}^2}\left(1 + \delta^{(i)}(\mathbf{q}^{(i)}) + \frac{\mathbf{p}\cdot\nabla_q\psi^{(i)}}{\bar{\sigma}^2}\right)\right]. \tag{7.10}$$

The only thing left to do is to eactually evaluate δq and δp, for this we need to evaluate the Zel'dovich propagator Ψ as well as the integra (reference integral here please). If everything is

done coorectly, we are left with an expression for f:

$$
\begin{aligned}
f \approx\ & \frac{\bar{\rho}(1+\delta^{(1)(i)}(q^{(i)}))}{(2\pi\bar{\sigma})^{3/2}} \exp\left(-\frac{p^2}{2\bar{\sigma}}\right)\left(1+\frac{p\cdot\nabla_q\psi^{(1)(i)}}{\bar{\sigma}}\right) \\
& + \frac{\bar{\rho}}{(2\pi\bar{\sigma})^{3/2}} \exp\left(-\frac{p^2}{2\bar{\sigma}}\right)\left(\nabla_q\delta^{(1)(i)}(q^{(i)}) + \frac{p\,\mathcal{H}[\psi^{(1)(i)}]}{\bar{\sigma}}\right) \\
& \quad \cdot \left\{\frac{4}{m}\left(\frac{1}{9}e^{3\eta} - \frac{16}{9}e^{3\eta/2} - \frac{9}{30}e^{2\eta} + \frac{1}{6} + \frac{2}{15}e^{-\eta/2} + \frac{11}{3}e^{\eta} - 2e^{\eta/2}\right) H^{(q)}_{\text{ini}} + \frac{4}{m}\left(\frac{2}{3}e^{\eta} - 2 + \frac{4}{3}e^{-\eta/2}\right)\mathbb{I}\right\} F_{\text{ini}} \\
& + \frac{\bar{\rho}}{(2\pi\bar{\sigma})^{3/2}} \exp\left(-\frac{p^2}{2\bar{\sigma}}\right)\left(\frac{\nabla_q\psi^{(1)(i)}}{\bar{\sigma}} - \frac{p}{\bar{\sigma}}\left(1+\delta^{(1)(i)}(q^{(i)})\right) + \frac{p\cdot\nabla_q\psi^{(1)(i)}}{\bar{\sigma}}\right) \\
& \quad \cdot \left\{2\left[\frac{12}{15}e^{2\eta} - \frac{4}{3}e^{\eta} - \frac{2}{15}e^{-\eta/2} + \frac{2}{3}e^{\eta/2}\right] H^{(i)}_{\text{ini}} + \left(\frac{2}{3}e^{\eta} - \frac{2}{3}e^{-\eta/2}\right)\mathbb{I}\right\} F_{\text{ini}}
\end{aligned}
\tag{7.11}
$$

CHAPTER 8

Conclusion